

Bones and their Markings

Axial Skeleton

Skull - cranial

Frontal bone – supraorbital foramen

Parietal bone (L&R)

Sphenoid bone – greater wings, lesser wings, foramen ovale, sella turcica

Temporal bone (L&R) – zygomatic process, mastoid process, external auditory/acoustic meatus, styloid

process, jugular foramen

Occipital bone – foramen magnum, occipital condyle, superior nuchal line, inferior nuchal line, external occipital protuberance

Ethmoid bone – crista galli, cribriform plate, perpendicular plate

Sutures – sagittal, coronal, lambdoid, squamous

Skull – facial

Maxillae (L&R) – palatine process, alveolar margin, infraorbital foramen

Lacrimal bones (L&R)

Nasal bones (L&R)

Palatine bones (L&R) – horizontal plates

Mandible – ramus, body, alveolar margin, mental foramen

Zygomatic bones (L&R) – temporal process

Vomer

Inferior nasal conchae (L&R)

Fetal Skull

Fontanelles – anterior, anterolateral (L & R), posterolateral (L & R), posterior

Hyoid bone (in throat above larynx)

Vertebrae

Spinous process

Body

Vertebral foramen

Superior and inferior articular facet

Superior and inferior articular process

Superior and inferior costal facet (thoracic only)

Transverse process

Transverse foramen (cervical only)

C1 – “atlas” – no body

C2 – “axis” – dens/odontoid process

Vertebrae – sacrum & coccyx

Sacrum - medial sacral crest, alae, sacral canal, lateral sacral crest, transverse line

Coccyx

Bony Thorax – sternum

Manubrium

Body

Xiphoid process

Bony Thorax – ribs

True/vertebrosternal (pairs 1-7)

False (pairs 8-12)

 Pairs 8-10 vertebrochondral

 Pairs 11 & 12 vertebral/floating

Appendicular Skeleton

Upper Extremities

Clavicle – sternal end, acromial end

Scapula – acromion, coracoid process, glenoid cavity, spine, medial border, lateral border, supraspinous fossa, infraspinous fossa

Humerus – head, greater tubercle, lesser tubercle, deltoid tuberosity, trochlea, capitulum, olecranon fossa, coronoid fossa, medial epicondyle, lateral epicondyle

Radius – head, radial tuberosity, styloid process

Ulna – olecranon, coronoid process, trochlear notch, head, styloid process

Hand

 8 carpals – scaphoid, lunate, triquetrum, pisiform, trapezium, trapezoid, capitate, hamate

 5 metacarpals

 14 phalanges – 5 proximal, 4 middle, 5 distal

Lower Extremities

Coxal bones

 Ilium – iliac crest, sacroiliac joint, greater sciatic notch, anterior superior iliac spine, anterior inferior iliac spine, posterior superior iliac spine, posterior inferior iliac spine

 Ischium – ischial spine, lesser sciatic notch

 Pubis – pubic symphysis

 Obturator foramen

 Acetabulum

Femur – head, neck, greater trochanter, lesser trochanter, linea aspera, medial condyle, lateral condyle, medial epicondyle, lateral epicondyle

Patella

Tibia – tibial tuberosity, medial malleolus

Fibula – head, lateral malleolus

Foot

 7 tarsals – talus, calcaneus, navicular, medial cuneiform, intermediate cuneiform, lateral cuneiform, cuboid

 5 metatarsals

 14 phalanges – 5 proximal, 4 middle, 5 distal